



SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS

November 2002

First HSC Assessment Task

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used
- All necessary working should be shown in every question

Total Marks – 60

- All Questions may be attempted
- Each Question is worth 12 marks

Examiner – *A.M. Gainford*

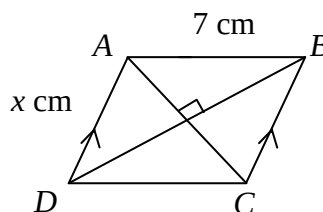
Question 1. (12 Marks) (Start a new booklet.)

- (a) Calculate $(5 \cdot 413698 \times 10^{12}) \div (2 \cdot 910064 \times 10^{17})$, giving your answer in scientific notation, correct to 6 significant figures. **1**

- (b) Calculate the probability of obtaining a total of 8 when two standard dice are rolled. **1**

- (c) Factorise completely $2x^2 + 2x - 12$. **1**

- (d) In the figure $ABCD$, $AB \parallel CD$, $AD \parallel CB$ and $AC \perp BD$. If $AB = 7\text{cm}$, $AD = x\text{cm}$, find the value of x . **1**



- (e) Write an equation for the parabola with vertex $(0, 0)$ and focus $(0, 2)$. **1**

- (f) Sketch on the number plane the graph of the function $y = \log_3 x$ in the domain $0 < x \leq 9$. **1**

- (g) Find the 24th term of the arithmetic series $7 + 4\frac{1}{2} + 2 + \dots$ **1**

- (h) Given that $\log_a 10 = 2 \cdot 094$ and $\log_a 2 = 0 \cdot 6309$, find $\log_a 500$. **1**

- (i) Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$. **2**

- (j) Find the values of k for which $x^2 + kx + 2$ is positive definite. **2**

Question 2. (12 Marks) (Start a new booklet.)

(a) Find and simplify the derivative of each of the following: 6

(i) $x^3 - 3x^2 + x - 2$

(ii) $\sqrt{1-x}$

(iii) $x\sqrt{1-x^2}$

(iv) $\frac{x+2}{1-x}$

(b) Consider the parabola with equation $y = \frac{1}{4}x^2 - x$ 4

(i) Write the equation in the form $(x-h)^2 = 4a(y-k)$.

(ii) State the vertex and focus.

(iii) Sketch the curve.

(c) (i) Evaluate $\sum_{r=1}^5 2r - 1$ 2

(ii) Express the geometric series $1 + 2 + 4 + 8 + \dots + 256$ in sigma notation.

Question 3. (12 Marks) (Start a new booklet.)

- (a) Seventy-five tagged fish are released into a dam known to contain fish. Later a sample of forty-two fish was netted from this dam and then released. Of these forty-two fish it was noted that five were tagged. **1**

Estimate the number of fish in the dam.

- (b) The vertices of a triangle are $A(3,4)$, $B(-2,2)$ and $C(5,-3)$. **8**

- (i) Find the coordinates of D , the midpoint of the side BC .
- (ii) Write down the equation of the side AB .
- (iii) Find the equation of the line through C parallel to AD .
- (iv) Find the coordinates of E , the point of intersection of the two lines described in parts (ii) and (iii).

- (c) (i) Find the value of m for which $\log(9^m) = \log 3 - \log \sqrt{3}$. **3**

- (ii) Evaluate $\log_b a \times \log_a b$.

Question 4. (12 Marks) (Start a new booklet.)

- (a) Given the expression $2x^2 + 4x - 1$: 2
- (i) Find the value of x when the expression has its minimum value.
- (ii) State the minimum value of this expression
- (b) Find the co-ordinates of the point on the curve $y = x^3 + 3x^2 + 3x - 7$ where the gradient of the tangent is zero. 2
- (c) The twelfth term of an arithmetic series is 2, and the fifteenth term is -4 . Find the first term and the common difference. 2
- (d) (i) Write a quadratic equation with roots -5 and 7 . 3
- (ii) Solve the quadratic inequality $x^2 - 2x - 3 \geq 0$.
- (e) Consider the recurring decimal fraction $F = 0.4\dot{3}\dot{7}$. 3
- (i) Express F as an infinite sum of terms, all but the first of which form a geometric series.
- (ii) Hence or otherwise express F as a common fraction in lowest terms.

Question 5. (12 Marks) (Start a new booklet.)

(a) Solve the equation $x^4 - 3x^2 + 2 = 0$. 2

(b) Consider the curve with equation $y = x - \frac{1}{x}$, $x > 0$: 3

- (i) Find the gradient of the tangent at the point on the curve where $x = 2$.
- (ii) Write in general form the equations of the tangent and normal to the curve at this point.

(c) An amount $\$A$ is borrowed at $r\%$ per annum reducible interest, calculated monthly. 7
The loan is to be repaid in equal monthly instalments of $\$M$.

Let $R = \left(1 + \frac{r}{1200}\right)$ and let $\$B_n$ be the amount owing after n monthly repayments have been made.

(i) Show that $B_n = AR^n - M \left(\frac{R^n - 1}{R - 1} \right)$.

Pat borrows $\$90\,000$ at 8% per annum reducible interest, calculated monthly. The loan is to be repaid in 96 equal monthly instalments.

- (ii) Show that the monthly repayments should be $\$1272 \cdot 30$.
- (iii) With the twenty-fourth payment, Pat pays an additional $\$10\,000$, so this payment is $\$11\,272 \cdot 30$. After this, repayments continue at $\$1272 \cdot 30$ per month. How many more repayments will be needed?

This is the end of the paper.



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Sample Solutions

Question 1

1) a) $5413698 \times 10^{12} \div 2.910064 \times 10^{17}$

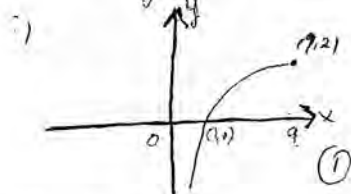
b.5f) 1.86037×10^{-5} (1)

b) $\frac{5}{36}$ (1)

c) $2(x^2 + x - 6)$ (1) *factorise completely*
 $2(x+3)(x-2)$

d) $x=7$ (1)

e) $x^2 = 8y$ (1)



g) $d = -2t$
 $a = 7$
 $T_{24} = a + 23d$
 $= 7 + 23(-2)$
 $= 7 - 46$
 $= -39$ (1)

h) $\log 500 = \log \frac{1000}{2}$
 $= \log 1000 - \log 2$
 $= 3 \log 10 - \log 2$
 $= 6.282 - 0.6309$
 $= 5.6511$ (3)

i) $\lim_{x \rightarrow 3} \frac{(x+3)(x-3)}{x-3} = \lim_{x \rightarrow 3} (x+3) = 6$ (2)

j) $k^2 - 8 < 0$
 $-\sqrt{8} < k < \sqrt{8}$

Q1.

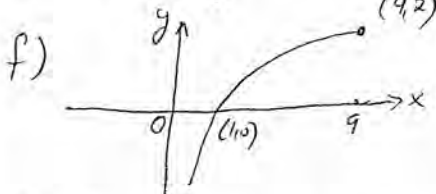
a) 1.86037×10^{-5}

b) $\frac{5}{36}$

c) $2(x+3)(x-2)$

d) $x=7$

e) $x^2 = 8y$



g) $T_{24} = a + 23d = -50$

h) 5.6511

i) 6 (2)

j) $-\sqrt{8} < k < \sqrt{8}$ (2)

Question 2

(a) (i) $\frac{d}{dx}(x^3 - 3x^2 + x - 2) = 3x^2 - 6x + 1$

(ii) $\sqrt{1-x} = (1-x)^{\frac{1}{2}}$
 $\frac{d}{dx}(1-x)^{\frac{1}{2}} = \frac{1}{2}(1-x)^{-\frac{1}{2}} \times -1$

$$= -\frac{1}{2}(1-x)^{-\frac{1}{2}}$$

$$= -\frac{1}{2\sqrt{1-x}}$$

(iii) $x\sqrt{1-x^2}$

$$u = x \quad v = \sqrt{1-x^2} = (1-x^2)^{\frac{1}{2}}$$

$$u' = 1 \quad v' = \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \times -2x = -x(1-x^2)^{-\frac{1}{2}} = -\frac{x}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(x\sqrt{1-x^2}) = vu' + uv'$$

$$= \sqrt{1-x^2} \times 1 + x \times -\frac{x}{\sqrt{1-x^2}}$$

$$= \sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}}$$

$$= \frac{1-x^2-x^2}{\sqrt{1-x^2}}$$

$$= \frac{1-2x^2}{\sqrt{1-x^2}}$$

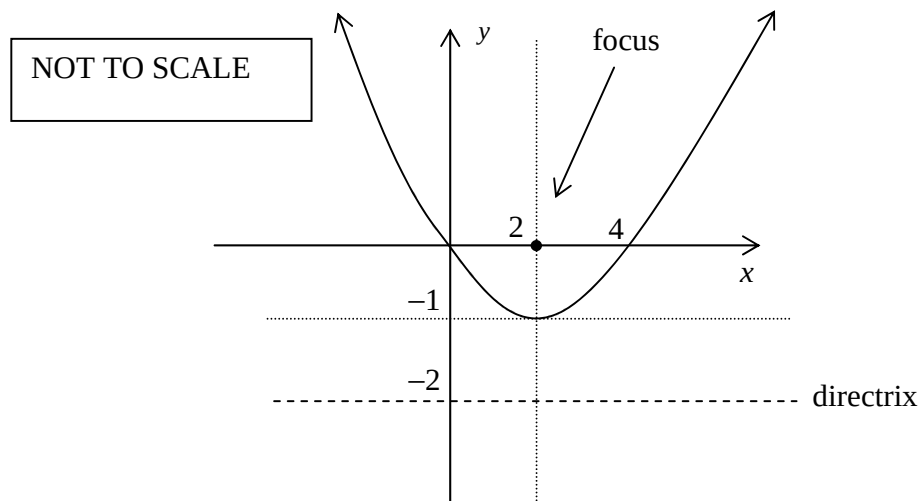
(iv) $\frac{x+2}{1-x}$

$$u = x+2 \quad v = 1-x$$

$$u' = 1 \quad v' = -1$$

$$\begin{aligned}
 \frac{d}{dx} \left(\frac{x+2}{1-x} \right) &= \frac{vu' - uv'}{v^2} \\
 &= \frac{(1-x) \times 1 - (x+2) \times -1}{(1-x)^2} \\
 &= \frac{1-x+x+2}{(1-x)^2} = \frac{3}{(1-x)^2} \\
 &= \frac{3}{(x-1)^2}
 \end{aligned}$$

- (b) (i) $y = \frac{1}{4}x^2 - x$
 $4y = x^2 - 4x$
 $4y + 4 = x^2 - 4x + 4$ (complete the square)
 $4(y+1) = (x-2)^2$
 $\therefore (x-2)^2 = 4(y+1)$
 $h=2, a=1, k=-1$
- (ii) Vertex (2, -1)
Focus (2, 0) (1 unit above the vertex)
- (iii)



- (c) (i) $\sum_{r=1}^5 (2r-1) = 1+3+5+7+9 = 25$
- (ii) $1+2+4+8+K+256 = \sum_{r=1}^9 2^{r-1} = \sum_{r=0}^8 2^r$

Question 3

a) $\frac{5}{42}$ of the sample were tagged. We assume a random distribution of fish so if x is the total number of fish then.

$$\frac{5}{42} = \frac{75}{x} \quad \therefore x = 75 \times \frac{42}{5} = \underline{\underline{630 \text{ fish}}}$$

b) i) $x_0 =$ is $\frac{5-2}{2}$ ie 1.5

y_0 is $\frac{-3+2}{2}$ ie -0.5

So coord of D is $\underline{\underline{\left(1\frac{1}{2}, -\frac{1}{2}\right)}}$

ii) $\frac{y-4}{x-3} = \frac{2-4}{-2-3} \Rightarrow -5(y-4) = -2(x-3)$

ie $-5y+20 = -2x+6$ OR $\underline{\underline{5y-2x-14=0}}$

ii) slope of AD is $\frac{4-(-\frac{1}{2})}{3-1\frac{1}{2}} = \frac{9 \times \frac{2}{3}}{2} = 3$

So required line is $\frac{y+3}{x-5} = 3$ ie $\underline{\underline{y=3x-18}}$

iv) Solve simultaneously, $5y-2x-14=0$ --- (1)
 $y-3x+18=0$ --- (2)

$5 \times (2) - (1)$ gives: $0-13x+104=0$
ie $x=104/13=8$

Sub $x=8$ into (1) or (2) gives.

$y=24-18=6$ so point of intersection is

$\underline{\underline{E(8,6)}}$

c) i) $\log q^n = \log 3 - \log \sqrt{3}$ find n .

Here the base of the logarithm is arbitrary so

lets use base 3. Raising eqn (1) to the power of 3 gives

$q^{3n} = 3 - \sqrt{3}$

ie $\left(\frac{3^2}{3}\right)^n = 3 - 3^{\frac{1}{2}}$

ie this is true iff $2n = \frac{1}{2}$

$\therefore n = \frac{1}{2} \div 2 = \frac{1}{4}$

ii) $\log_b a \times \log_a b$ using the change of base rule.

$\log_b a = \frac{\log a}{\log b}$ ie sub this into (1) gives

$\frac{\log a}{\log a} \times \log_a b = 1$

Question 4

$$(a) \quad 2x^2 + 4x - 1$$

$$\text{axis } x = \frac{-b}{2a}$$

$$x = -1 \quad y = -3$$

$$(ii) \quad x = -1$$

$$(ii) \quad y = -3$$

$$(b) \quad y' = 3x^2 + 6x + 3$$

$$3x^2 + 6x + 3 = 0$$

$$x^2 + 2x + 1 = 0$$

$$(x+1)(x+1) = 0$$

$$\left[\begin{array}{l} x = -1 \\ y = -8 \end{array} \right] y = (-1)^3 + 3(-1)^2 + 3(-1) - 7$$

$$c) \quad a + 11d = 2$$

$$\textcircled{2} \quad a + 14d = -4$$

$$\textcircled{3} \quad 3d = -6$$

$$d = -2$$

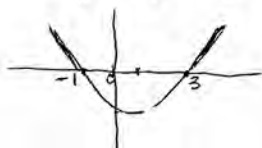
$$\textcircled{1} \quad a - 22 = 2$$

$$\text{Hence } a = 24$$

$$x) \quad (x+5)(x-7) = 0$$

$$(ii) \quad x^2 - 2x - 3 \geq 0$$

$$(x-3)(x+1) \geq 0$$



$$x \leq -1 \text{ or } x \geq 3$$

$$(c) (i) F = 0.4 + [0.037 + 0.00037 \dots]$$

(ii) after 1st term geometric series

$$a = 0.037 \text{ or } \frac{37}{1000}$$

$$r = \frac{1}{100}$$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{\frac{37}{1000}}{1 - \frac{1}{100}}$$

$$= \frac{37}{1000} \times \frac{100}{99}$$

$$= \frac{37}{990}$$

$$F = \frac{4}{10} + \frac{37}{990}$$

$$= \frac{433}{990}$$

or

$$x = 0.4373737 \dots$$

$$10x = 4.373737 \dots$$

$$1000x = 437.3737 \dots$$

$$990x = 433$$

$$x = \frac{433}{990}$$

Question 5

$$(a) \quad x^4 - 3x^2 + 2 = 0$$

$$(x^2 - 1)(x^2 - 2) = 0$$

So $x^2 - 1 = 0$ and $x^2 - 2 = 0$

$$x^2 = 1 \quad x^2 = 2$$

$$x = \pm 1 \quad (i) \quad x = \pm \sqrt{2} \quad (i)$$

$$(b) \quad y = x - \frac{1}{x}$$

$$y = x - x^{-1}$$

$$\frac{dy}{dx} = 1 + x^{-2} = 1 + \frac{1}{x^2}$$

$$(i) \text{ at } x=2, m = 1 + \frac{1}{2^2} = 1 + \frac{1}{4} = \frac{5}{4} \quad (i)$$

$$(ii) \text{ At } x=2, y = 2 - \frac{1}{2} = \frac{3}{2} \text{ point } (2, \frac{3}{2})$$

$$\begin{array}{l} \text{gradient tangent } \frac{5}{4} \\ \text{gradient normal } -\frac{4}{5} \end{array}$$

$$\text{eq}^n \text{ tangent } (y - \frac{3}{2}) = \frac{5}{4}(x - 2)$$

$$4y - 6 = 5x - 10$$

$$\therefore 5x - 4y - 4 = 0 \quad (i)$$

$$\text{eq}^n \text{ normal } (y - \frac{3}{2}) = -\frac{4}{5}(x - 2)$$

$$5y - \frac{15}{2} = -4x + 8$$

$$10y - 15 = -8x + 16$$

$$\therefore 8x + 10y - 31 = 0 \quad (i)$$

(c)(i) \$A\$ at $r\%$ p.a. calculated monthly -
equal instalments \$M\$.

$$\$B_1 = A + \left(A \times \frac{r\%}{12} \right) - M$$

$$= A + \frac{Ar}{1200} - M$$

$$= A \left(1 + \frac{r}{1200} \right) - M$$

$$\$B_2 = \left[A \left(1 + \frac{r}{1200} \right) - M \right] + \left[A \left(1 + \frac{r}{1200} \right) - M \right] \times \frac{r}{1200} - M$$

$$= \left[A \left(1 + \frac{r}{1200} \right) - M \right] \left[1 + \frac{r}{1200} \right] - M$$

$$= A \left(1 + \frac{r}{1200} \right)^2 - \left(1 + \frac{r}{1200} \right) M - M$$

$$= A \left(1 + \frac{r}{1200} \right)^2 - M \left[1 + \left(1 + \frac{r}{1200} \right) \right]$$

$$= A(R)^2 - M(1+R)$$

$$\therefore \$B_n = AR^n - M(1+R+\dots+R^{n-1})$$

$$\text{using } S_n = \frac{R^n - 1}{R - 1} = \frac{R \times R^{n-1} - 1}{R - 1} = \frac{R^n - 1}{R - 1}$$

$$\therefore \$B_n = AR^n - M \left(\frac{R^n - 1}{R - 1} \right) \text{ as required. } (3)$$

(c) (i) \$90,000 $R = 1.006 = 1 \frac{1}{150}$ 96 instalments

$$0 = 90,000 \times \left(1 \frac{1}{150}\right)^{96} - m \left(\frac{\left(1 \frac{1}{150}\right)^{96} - 1}{\left(1 \frac{1}{150}\right) - 1}\right)$$

$$m = \frac{90,000 \times \left(1 \frac{1}{150}\right)^{96}}{\left(\frac{\left(1 \frac{1}{150}\right)^{96} - 1}{\left(1 \frac{1}{150}\right) - 1}\right)} = \$1272.30 \quad (1)$$

$$(ii) \$B_{24} = 90,000 \times \left(1 \frac{1}{150}\right)^{24} - 1272.30 \left(\frac{\left(1 \frac{1}{150}\right)^{24} - 1}{\left(1 \frac{1}{150}\right) - 1}\right)$$

$$= \$72565.1165$$

Now $\$B_{24} = \$72565.1165 - \$10,000 = \62565.1165

is yet to be paid.

$$\text{So } 0 = 62565.1165 \times \left(1 \frac{1}{150}\right)^n - 1272.30 \left(\frac{\left(1 \frac{1}{150}\right)^n - 1}{\left(1 \frac{1}{150}\right) - 1}\right)$$

$$0 = 62565.1165 \times 1.006^n - 190845 \left(1.006^n - 1\right)$$

$$0 = 62565.1165 \times 1.006^n - 190845 \times 1.006^n + 190845$$

$$-190845 = -128279.8835 \times 1.006^n$$

$$1.487723521 = 1.006^n$$

$$\ln 1.487723521 = n \ln 1.006$$

$$n = \frac{\ln 1.487723521}{\ln 1.006}$$

$$n = 59.785$$

$\Rightarrow$ 60 more payments (3)